task 7

Using only AWS CLI:

Create a custom VPC
Create 2 public subnets (pub/pvt)
Create Internet Gateway
Create NAT Gateway
Create Route Table
Create Security Group
Launch 2 EC2 instances (1 in each Subnet)
Install Nginx via user data
Create Application Load Balancer
Register instances
Test via ALB DNS

steps :

vpc-0dbd627019453a185

"InternetGatewayId": "igw-0a27ff5f767b5ee79”

pub - "SubnetId": "subnet-0ba0c8dfc031654d4",

pub subnet - 2 : subnet-01a639b22e440a07e
pvt - "SubnetId": "subnet-0197111928757732d”

"RouteTableId": "rtb-08caebc737a955a21",

"AllocationId": "eipalloc-0c295f4318a4a9dcf",

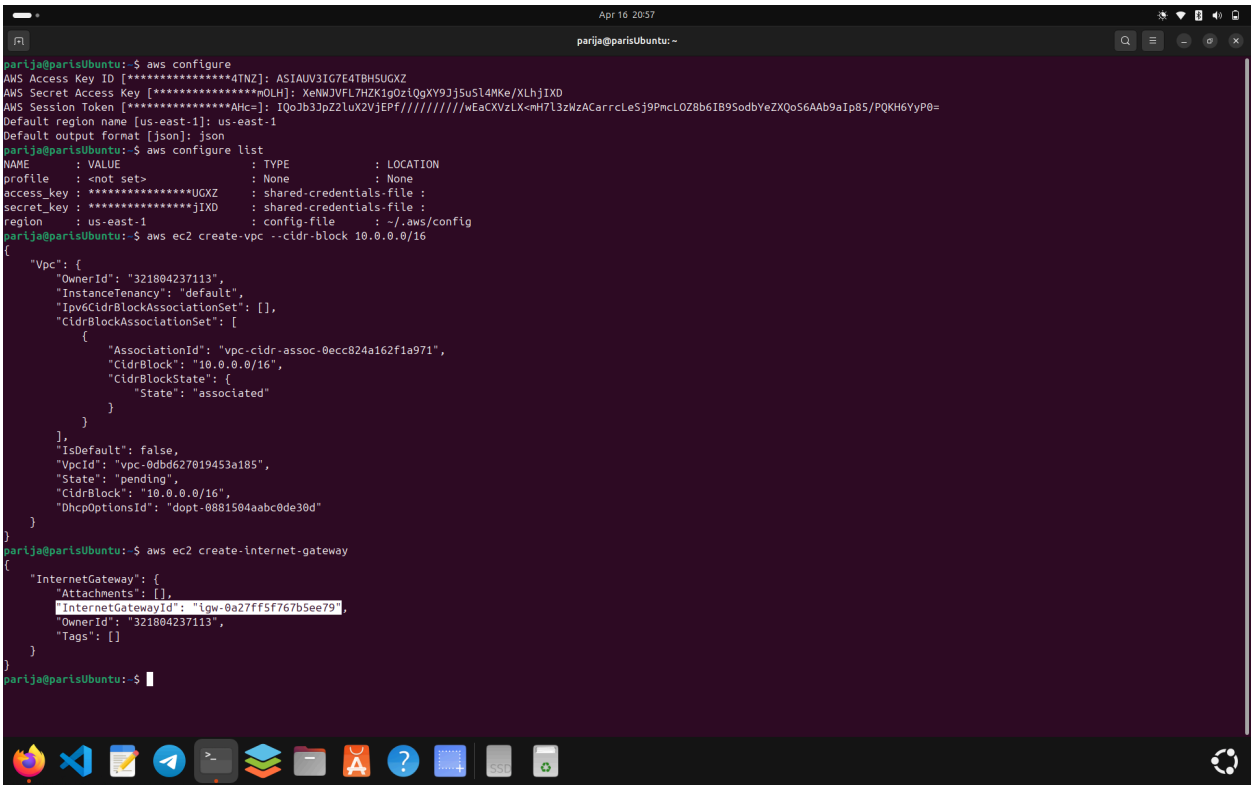
"NatGatewayId": "nat-08bd4e8e67f0b2441",

"RouteTableId": "rtb-0fe5d696496125c84",

"GroupId": "sg-0168c0c3db6376065",
ec2 - pub "InstanceId": "i-02c831f2dfdf2ad4c",
ec2 -pvt "InstanceId": "i-0e3a83d4ffb5a1b0b",

security sg = subnet-0ba0c8dfc031654d4

"TargetGroupArn": "arn:aws:elasticloadbalancing:us-east-1:321804237113:targetgroup/my-tg/6399f94f5d90d72a",



```
parija@parisUbuntu:~$ aws ec2 attach-internet-gateway --vpc-id vpc-0dbd627019453a185 --interne
t-gateway-id igw-0a27ff5f767b5ee79
parija@parisUbuntu:~$ aws ec2 create-subnet --vpc-id vpc-0dbd627019453a185 --cidr-block 10.0.1
.0/24 --availability-zone us-east-1a
{
    "Subnet": {
        "AvailabilityZoneId": "use1-az6",
        "MapCustomerOwnedIpOnLaunch": false,
        "OwnerId": "321804237113",
        "AssignIpv6AddressOnCreation": false,
        "Ipv6CidrBlockAssociationSet": [],
        "SubnetArn": "arn:aws:ec2:us-east-1:321804237113:subnet/subnet-0ba0c8dfc031654d4",
        "EnableDns64": false,
        "Ipv6Native": false,
        "PrivateDnsNameOptionsOnLaunch": {
            "HostnameType": "ip-name",
            "EnableResourceNameDnsARecord": false,
            "EnableResourceNameDnsAAAARecord": false
        },
        "SubnetId": "subnet-0ba0c8dfc031654d4",
        "State": "available",
        "VpcId": "vpc-0dbd627019453a185",
        "CidrBlock": "10.0.1.0/24",
        "AvailableIpAddressCount": 251,
        "AvailabilityZone": "us-east-1a",
        "DefaultForAz": false,
        "MapPublicIpOnLaunch": false
    }
}
parija@parisUbuntu:~$
```

```
parija@parisUbuntu:~$ aws ec2 allocate-address
{
    "AllocationId": "eipalloc-0c295f4318a4a9dcf",
    "PublicIpv4Pool": "amazon",
    "NetworkBorderGroup": "us-east-1",
    "Domain": "vpc",
    "PublicIp": "44.209.231.15"
}
parija@parisUbuntu:~$ aws ec2 create-nat-gateway --subnet-id subnet-0ba0c8dfc031654d4 --alloca
tion-id eipalloc-0c295f4318a4a9dcf
{
    "ClientToken": "9daa7a3d-8a55-4439-b550-0848d391b339",
    "NatGateway": {
        "CreateTime": "2026-04-16T16:42:44+00:00",
        "NatGatewayAddresses": [
            {
                "AllocationId": "eipalloc-0c295f4318a4a9dcf",
                "IsPrimary": true,
                "Status": "associating"
            }
        ],
        "NatGatewayId": "nat-08bd4e8e67f0b2441",
        "State": "pending",
        "SubnetId": "subnet-0ba0c8dfc031654d4",
        "VpcId": "vpc-0dbd627019453a185",
        "ConnectivityType": "public",
        "AvailabilityMode": "zonal"
    }
}
parija@parisUbuntu:~$ ^C
parija@parisUbuntu:~$ aws ec2 create-route-table --vpc-id vpc-0dbd627019453a185
{
    "RouteTable": {
        "Associations": [],
        "PropagatingVgws": [],
        "RouteTableId": "rtb-0fe5d696496125c84",
        "Routes": [
            {
                "DestinationCidrBlock": "10.0.0.0/16",
                "GatewayId": "local",
                "Origin": "CreateRouteTable",
                "State": "active"
            }
        ],
        "Tags": [],
        "VpcId": "vpc-0dbd627019453a185",
        "OwnerId": "321804237113"
    }
```

adding security group

```
parija@parisUbuntu:~$ aws ec2 create-security-group --group-name my-sg --description "allow http and ssh" --vpc-id vpc-0dbd627019453a185
{
    "GroupId": "sg-0168c0c3db6376065",
    "SecurityGroupArn": "arn:aws:ec2:us-east-1:321804237113:security-group/sg-0168c0c3db6376065"
}
parija@parisUbuntu:~$ aws ec2 authorize-security-group-ingress --group-id sg-0168c0c3db6376065 --protocol tcp --port 22 --cidr 0.0.0.0/0
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-03db494055f3a8327",
            "GroupId": "sg-0168c0c3db6376065",
            "GroupOwnerId": "321804237113",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 22,
            "ToPort": 22,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:321804237113:security-group-rule/sgr-03db494055f3a8327"
        }
    ]
}
parija@parisUbuntu:~$ aws ec2 authorize-security-group-ingress --group-id sg-0168c0c3db6376065 --protocol tcp --port 80 --cidr 0.0.0.0/0
{
    "Return": true,
    "SecurityGroupRules": [
        {
            "SecurityGroupRuleId": "sgr-0bc5af432683ab09a",
            "GroupId": "sg-0168c0c3db6376065",
            "GroupOwnerId": "321804237113",
            "IsEgress": false,
            "IpProtocol": "tcp",
            "FromPort": 80,
            "ToPort": 80,
            "CidrIpv4": "0.0.0.0/0",
            "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:321804237113:security-group-rule/sgr-0bc5af432683ab09a"
        }
    ]
}
parija@parisUbuntu:~$
```

```
parija@parisUbuntu:~$ aws ec2 run-instances --image-id ami-0c02fb55956c7d316 --instance-type t2.micro --count 1 --key-name task7ec2 --security-group-ids sg-0168c0c3db6376065 --subnet-id subn
et-0ba0c8dfc031654d4 --user-data file://data.sh
{
    "ReservationId": "r-06ecec472e3c2a8d0",
    "OwnerId": "321804237113",
    "Groups": [],
    "Instances": [
        {
            "Architecture": "x86_64",
            "BlockDeviceMappings": [],
            "ClientToken": "f6bb90a3-2214-4e52-9ebc-e73effcf9f30",
            "EbsOptimized": false,
            "EnaSupport": true,
            "Hypervisor": "xen",
            "NetworkInterfaces": [
                {
                    "Attachment": {
                        "AttachTime": "2026-04-16T17:24:14+00:00",
                        "AttachmentId": "eni-attach-0240ef96bb5f204b2",
                        "DeleteOnTermination": true,
                        "DeviceIndex": 0,
                        "Status": "attaching",
                        "NetworkCardIndex": 0
                    },
                    "Description": "",
                    "Groups": [
                        {
                            "GroupId": "sg-0168c0c3db6376065",
                            "GroupName": "my-sg"
                        }
                    ],
                    "Ipv6Addresses": [],
                    "MacAddress": "0e:f2:27:8b:56:8b",
                    "NetworkInterfaceId": "eni-05238e3c4d3c51751",
                    "OwnerId": "321804237113",
                    "PrivateIpAddress": "10.0.1.233",
                    "PrivateIpAddresses": [
                        {
                            "Primary": true,
                            "PrivateIpAddress": "10.0.1.233"
                        }
                    ],
                    "SourceDestCheck": true,
                    "Status": "in-use",
                    "SubnetId": "subnet-0ba0c8dfc031654d4",
                    "VpcId": "vpc-0dbd627019453a185",
                    "InterfaceType": "interface",
```

before making alb - you will need a third public subnet

```
parija@parisUbuntu:~$ aws ec2 modify-subnet-attribute --subnet-id subnet-01a639b22e440a07e --m
ap-public-ip-on-launch
parija@parisUbuntu:~$ aws ec2 associate-route-table --subnet-id subnet-01a639b22e440a07e --rou
te-table-id rtb-08caebc737a955a21
{
    "AssociationId": "rtbassoc-0e690dcc1d087454f",
    "AssociationState": {
        "State": "associated"
    }
}
parija@parisUbuntu:~$ aws elbv2 create-load-balancer --name myalb --subnets subnet-0ba0c8dfc03
1654d4 subnet-01a639b22e440a07e --security-groups sg-0168c0c3db6376065
{
    "LoadBalancers": [
        {
            "LoadBalancerArn": "arn:aws:elasticloadbalancing:us-east-1:321804237113:loadbalanc
er/app/myalb/4d12ab0a229f576f",
            "DNSName": "myalb-963423018.us-east-1.elb.amazonaws.com",
            "CanonicalHostedZoneId": "Z35SXDOTRQ7X7K",
            "CreatedTime": "2026-04-16T17:40:02.409000+00:00",
            "LoadBalancerName": "myalb",
            "Scheme": "internet-facing",
            "VpcId": "vpc-0dbd627019453a185",
            "State": {
                "Code": "provisioning"
            },
            "Type": "application",
            "AvailabilityZones": [
                {
                    "ZoneName": "us-east-1b",
                    "SubnetId": "subnet-01a639b22e440a07e",
                    "LoadBalancerAddresses": []
                },
                {
                    "ZoneName": "us-east-1a",
                    "SubnetId": "subnet-0ba0c8dfc031654d4",
                    "LoadBalancerAddresses": []
                }
            ],
            "SecurityGroups": [
                "sg-0168c0c3db6376065"
            ],
            "IpAddressType": "ipv4"
        }
    ]
}
parija@parisUbuntu:~$
```

```
arija@parisUbuntu:~$ aws elbv2 create-listener \
 --load-balancer-arn arn:aws:elasticloadbalancing:us-east-1:321804237113:loadbalancer/app/myalb/4d12ab0a229f576f \
 --protocol HTTP \
 --port 80 \
 --default-actions Type=forward,TargetGroupArn=arn:aws:elasticloadbalancing:us-east-1:321804237113:targetgroup/my-tg/6399f94f5d90d72a

   "Listeners": [
       {
           "ListenerArn": "arn:aws:elasticloadbalancing:us-east-1:321804237113:listener/app/myalb/4d12ab0a229f576f/e6412af25d265ef5",
           "LoadBalancerArn": "arn:aws:elasticloadbalancing:us-east-1:321804237113:loadbalancer/app/myalb/4d12ab0a229f576f",
           "Port": 80,
           "Protocol": "HTTP",
           "DefaultActions": [
               {
                   "Type": "forward",
                   "TargetGroupArn": "arn:aws:elasticloadbalancing:us-east-1:321804237113:targetgroup/my-tg/6399f94f5d90d72a",
                   "ForwardConfig": {
                       "TargetGroups": [
                           {
                               "TargetGroupArn": "arn:aws:elasticloadbalancing:us-east-1:321804237113:targetgroup/my-tg/6399f94f5d90d72a",
                               "Weight": 1
                           }
                       ],
                       "TargetGroupStickinessConfig": {
                           "Enabled": false
                       }
                   }
               }
           ]
       }
   ]

arija@parisUbuntu:~$
```